

Learning for Autonomous Vehicles

– introduction

TSFS12: Autonomous Vehicles – planning, control, and learning systems

Lecture 10: Erik Frisk <erik.frisk@liu.se>

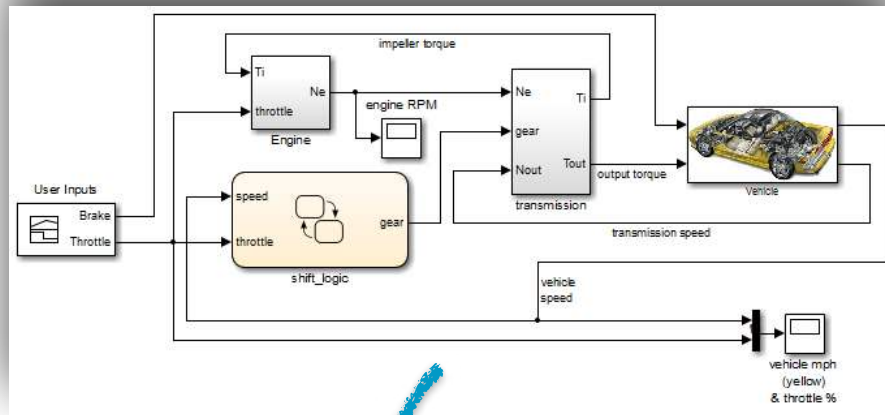
Course announcements

- Introduction to mini-projects on Tuesday, October 1, 08:15
Very important that you participate!
 - Groups of about 5 people in each group
- Guest lecturers October 17, 13:15
 - Hanna Pelikan, IKOS/LiU - how autonomous robots and humans interact
 - Einride
- HI4 now available on the course git-repo

Learning, models, and data

Q: What is (machine) learning?

A: using data to build models that can predict and/or act upon the world



predictive/control models

Learning and Autonomous Vehicles

- This is not a course in machine learning
- Challenging from a course perspective
 - some of you have no previous experience
 - some of you have extensive ML experience
- Objectives of this part of the course
 - Discuss learning in the context of autonomous vehicles
 - Identify key areas of ML that is interesting, recognize terms and context
 - Get some basic, hands-on, experience with basic methods
- If you want to dig deeper/know more: mini-project is an excellent opportunity
- In particular, two methods/models/areas will be discussed in some more detail
 - Neural Networks + Reinforcement learning

Reflections on the Learning-to-Control Renaissance

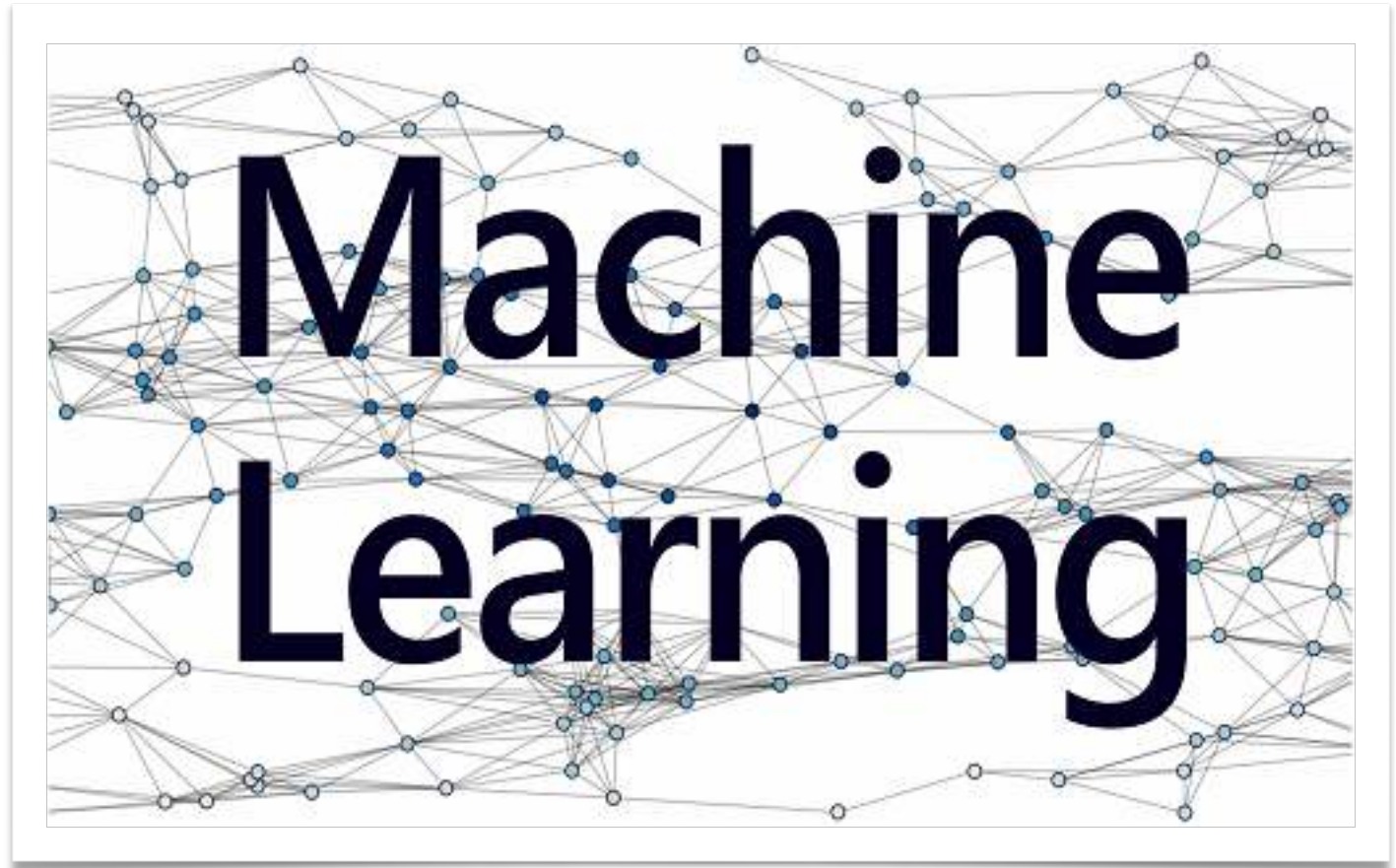
- Plenary Talk from the June 2020 IFAC World Congress (IFAC - International Federation of Automatic Control)
- Benjamin Recht - UC Berkley
- Slightly advanced — but highly recommended 45 minutes
- Modern discussion on learning, control, when they apply and how they can fit together
- A critical eye on RL and some constructive ideas forward
- As you'll see, I borrowed from this lecture a little



<https://youtu.be/IEZFwh8sw8s>

Rough characterization of machine learning tasks

- Supervised learning
 - Learn by examples
- Unsupervised learning
 - No labeled data
- Reinforcement learning
 - Learn without a model but with a reward function

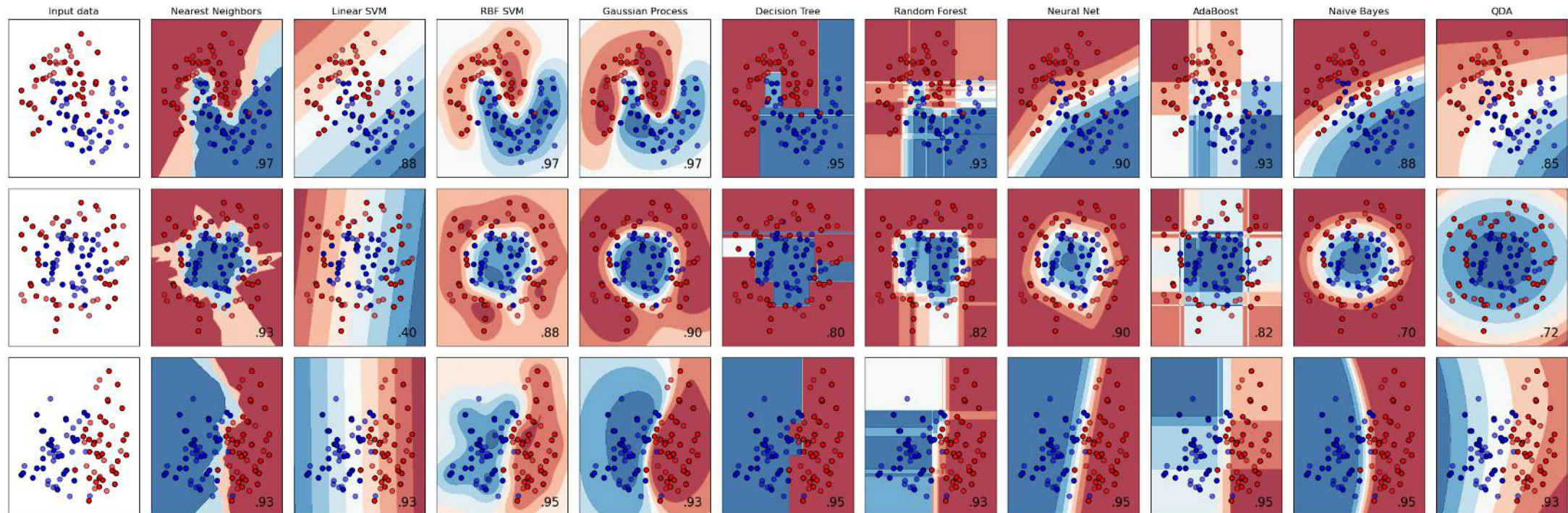


Learning and Autonomous Vehicles

- From the start, it became important to know where you are in the world
 - ▢▢▢▢➔ Localization and mapping
- Then it became important to know more about surroundings
 - ▢▢▢▢➔ Perception
 - Computer Vision
 - Sensor development, Lidar technology, ...
- A current hot topic is how to model and *predict* behavior of the environment
- Modeling camera inputs and agent behavior in an uncertain and complex world is difficult

data and learning an exciting
possibility going forward

Learning - some basics



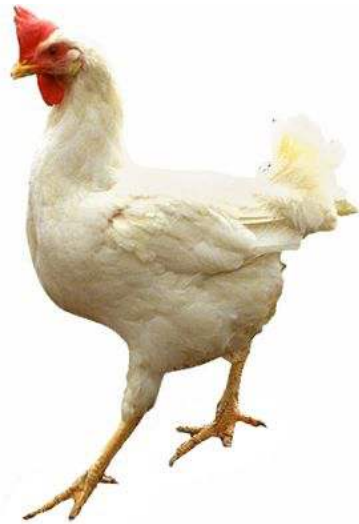
- Methods look for patterns/features in data to make classifiers or predictors
- Models often opaque — be careful what your model predicts
- Can be *very* sensitive when extrapolating into areas not covered by data

Ensuring safety and robustness with opaque models⁹

- Passenger cars
 - Increased traffic efficiency
 - New services
 - Safety — what can we expect? Rough calculation — 1% AV in US with human capabilities ~ more than 1 death/day involving AV¹
- For example aerospace industry has well developed models for validating safety and robustness
- How to validate these numbers with ML models where we “do not understand” how the models work internally?
- Big research and technology challenge going forward - certainly not yet solved

Edge cases are surprising and rare

This is not a human



or is it ...

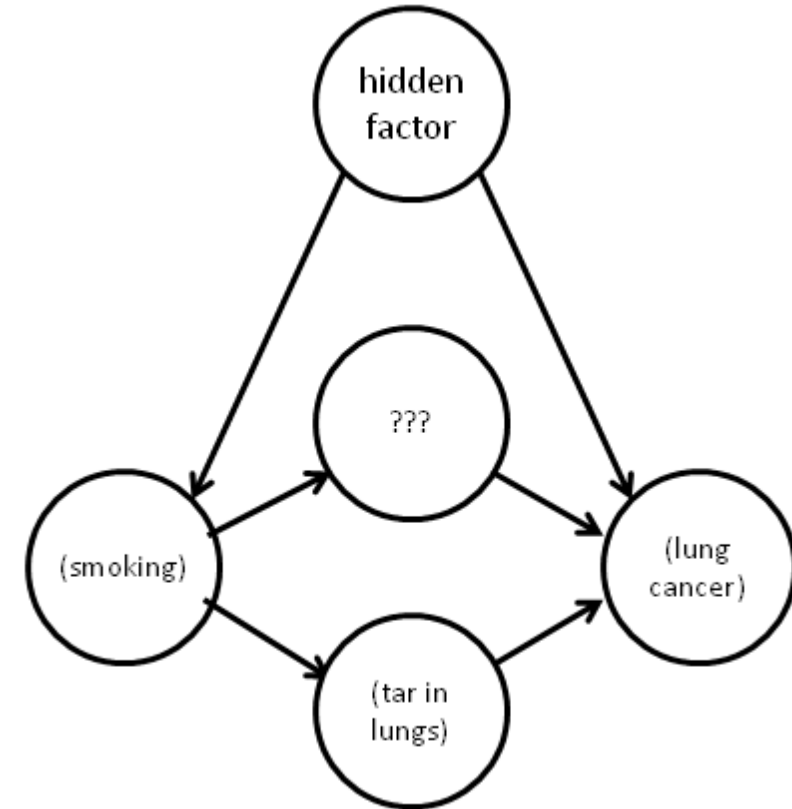


- You won't see them in testing; things no-one thought about
- Safety is a huge topic in autonomous systems
- Typical solution — an on-line system supervising methods and algorithms that are difficult to monitor, e.g.,

Wabersich, Kim Peter, and Melanie N. Zeilinger. "*A predictive safety filter for learning-based control of constrained nonlinear dynamical systems.*" *Automatica* 129 (2021)

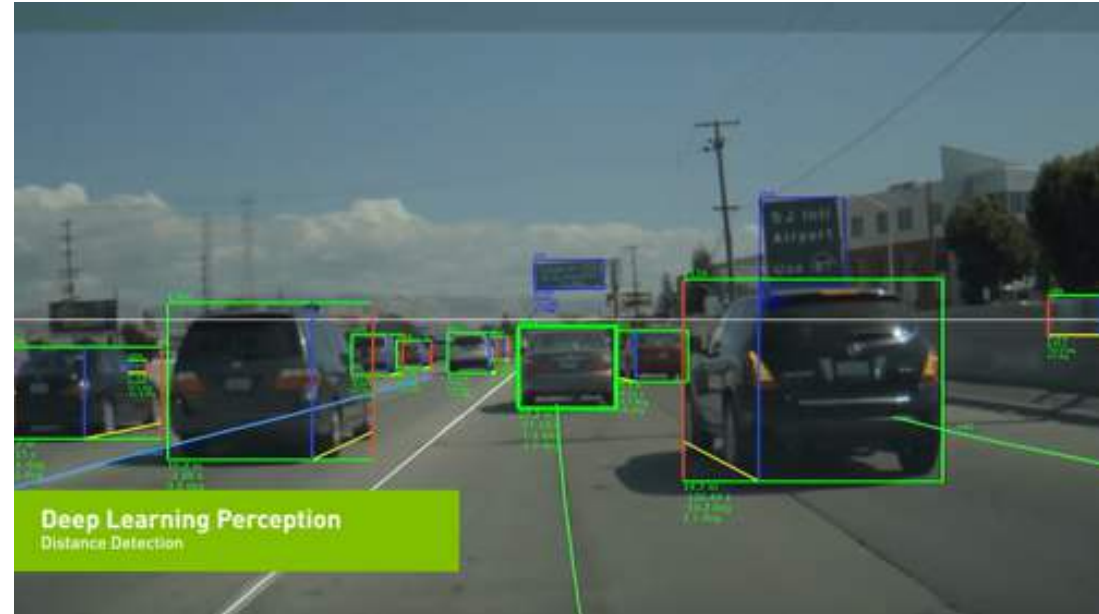
Learning - some basics

- Basic machine learning looks for patterns in data
 - Correlation models
 - It does not find cause and effect
- Extrapolation becomes brittle
- Models are only as good as your data
 - You need a lot of information (not the same as many data points)
- Look out for biased models (mathematically, socially, ethically, ...)



AI and machine learning

- Many current systems are hand-crafted and are heavily based on advanced sensor techniques
- Learning systems have the potential to make high-level autonomy a reality
- Probable components where deep-learning systems will be core
 - computer vision, situation awareness
 - model-based reinforcement learning for decision making
- ML will most likely be an important part of the solution (but most likely not *the only* solution)



Learning for Autonomous Vehicles

Where are we now and where are we learning?

14



Boston Dynamics

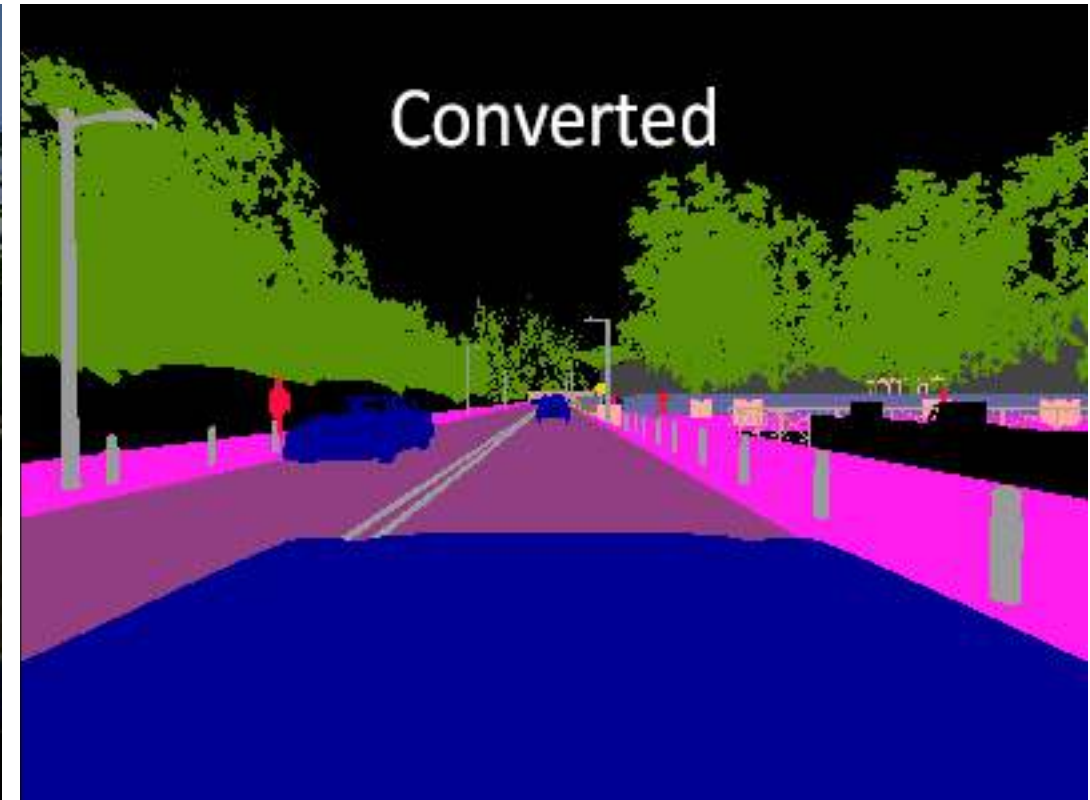


ETH Zürich



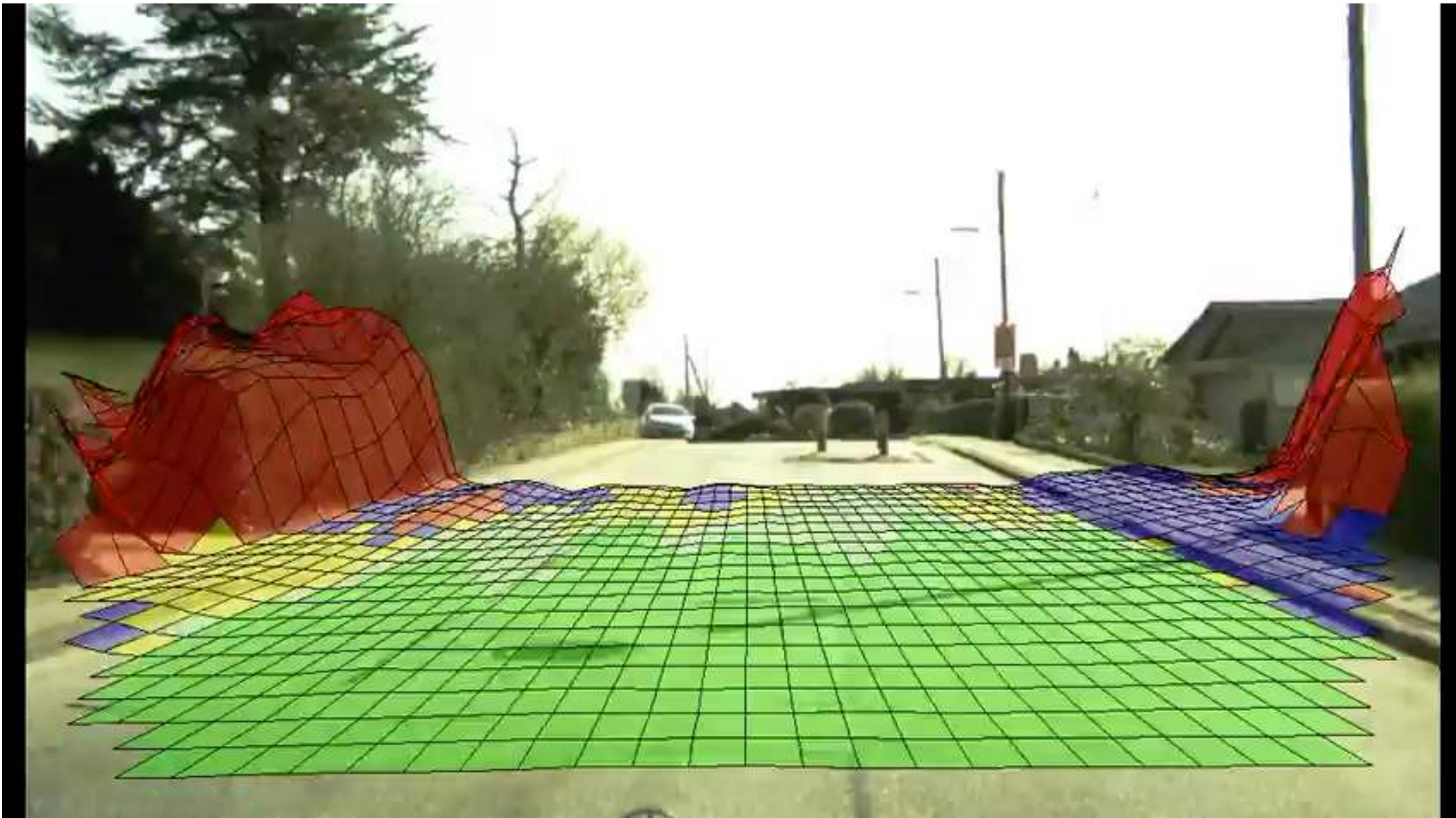
Waymo, Tesla, Cruise ...

Semantic segmentation in the CARLA simulator



Semantic segmentation for real-world data





Vision based systems

- Maps are good; but need detailed information about environment that is not included in maps,
 - Detection of other vehicles, pedestrians, cyclists,
 - Road boundaries, road condition, rain, free space ...
 - Traffic lights, signs, ...
 - Detection of people in a search and rescue mission
- Vision systems and cameras are (currently) cost-effective compared to, e.g., Lidar systems
 - Are Lidars essential or can they be replaced by learning-powered Vision systems?
- Learning techniques are core in these tasks

Vision based systems

- Machine learning methods have enhanced the field tremendously during the last years, but still a long way to go
 - Robustness
 - Reduce need for labeled data
 - Verification of function based on ML models
- Big area for autonomous vehicles that is not covered in this course: recommend courses from computer vision laboratory:

TSBB19 - Machine Learning for Computer Vision

Perception yes, but can we learn how to act?

- Reinforcement Learning (RL) is one possibility
- RL is learning what to do by *maximizing a reward* without being told what to do.
- Instead perform actions and evaluate the results and improve the policy/controller
- Discover actions on its own, by exploring and evaluating the outcome.
- Learn
 - policy/controller, $u_t = f(x_t)$
 - cost-to-go

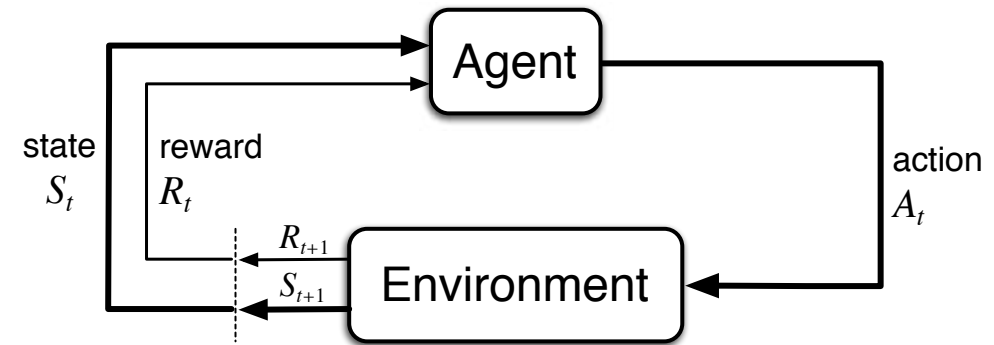
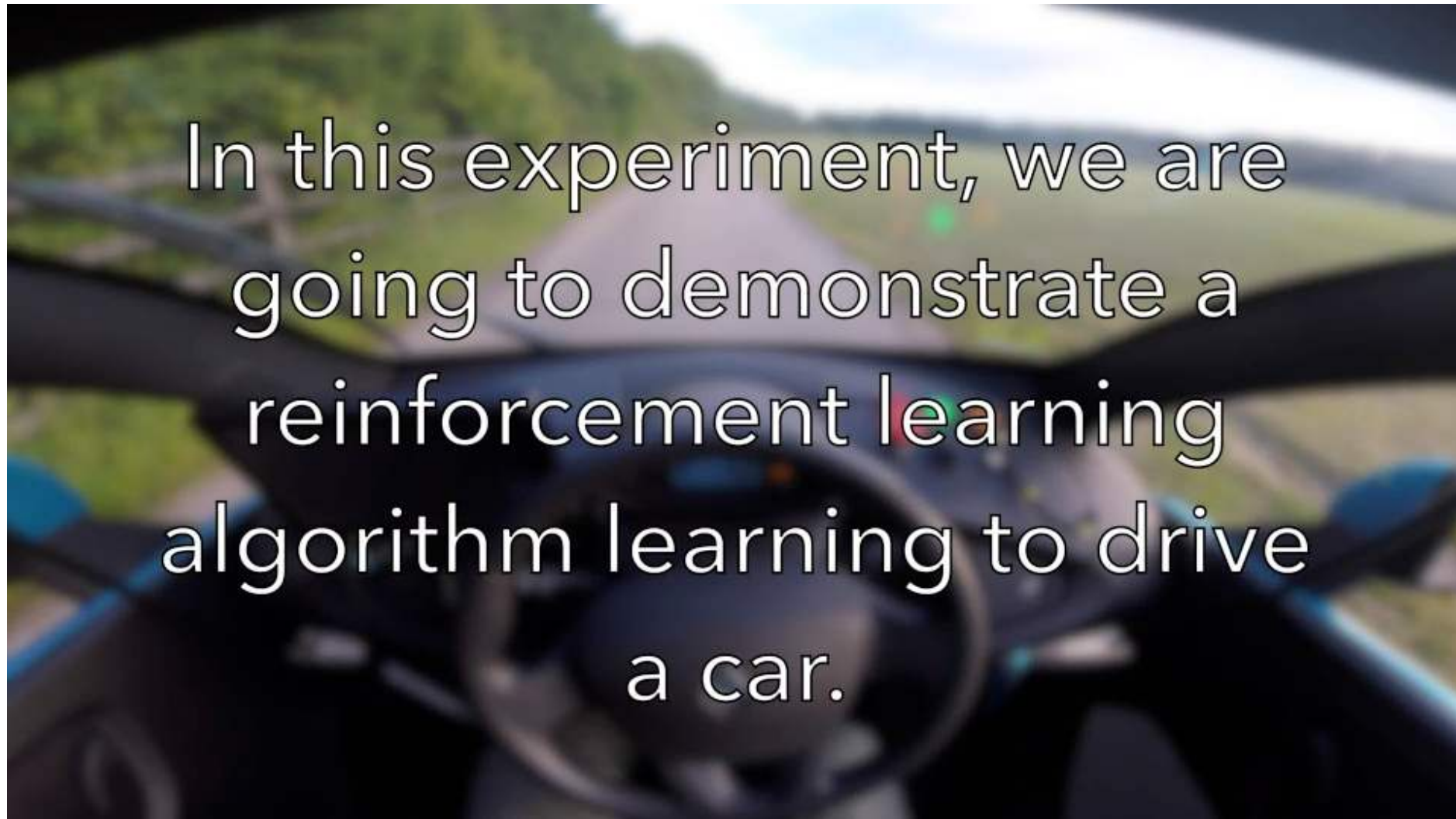


Figure 3.1: The agent–environment interaction in a Markov decision process.

“Reinforcement Learning: An Introduction”
Richard S. Sutton and Andrew G. Barto

End-to-end (camera-to-control) learning



Currently a hot topic ... in research and industry

IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, VOL. 23, NO. 6, JUNE 2022

Deep Reinforcement Learning for Autonomous Driving

IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, VOL. 23, NO. 6, JUNE 2022

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIV.2022.3167271, IEEE

JOURNAL OF LATEX CLASS FILES

Transactions on Intelligent Vehicles

1

A Receding Horizon Approach for

Xinglong Zhang,

Abstract—Kinodynamic motion
autonomous vehicles with high ma
vironments. However, obtaining a

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIV.2022.3165178, IEEE

Transactions on Intelligent Vehicles

1

Robust Lane Change Decision Making for Autonomous Vehicles: An Observation Adversarial Reinforcement Learning Approach

Xiangkun He, *Member, IEEE*, Haohan Yang, Zhongxu Hu, and Chen Lv, *Senior Member, IEEE*

End-to-end learning, deep reinforcement learning

- The result from the last video is kind of nice, but is it a good idea for real-life autonomous vehicles? It highlights some striking problems.
- If perception can give us road boundaries, do we need to learn how to drive at the center of the road?
- There are plenty of arguments
 - Learn to adapt to uncertainty
 - Model the world is too difficult, measure instead
 - Why should we learn things we already know
 - Use RL for high-level reasoning
 - ...
- Fair to say the jury is still out on this one.
- I highly recommend the Ben Recht plenary talk video for a deeper discussion

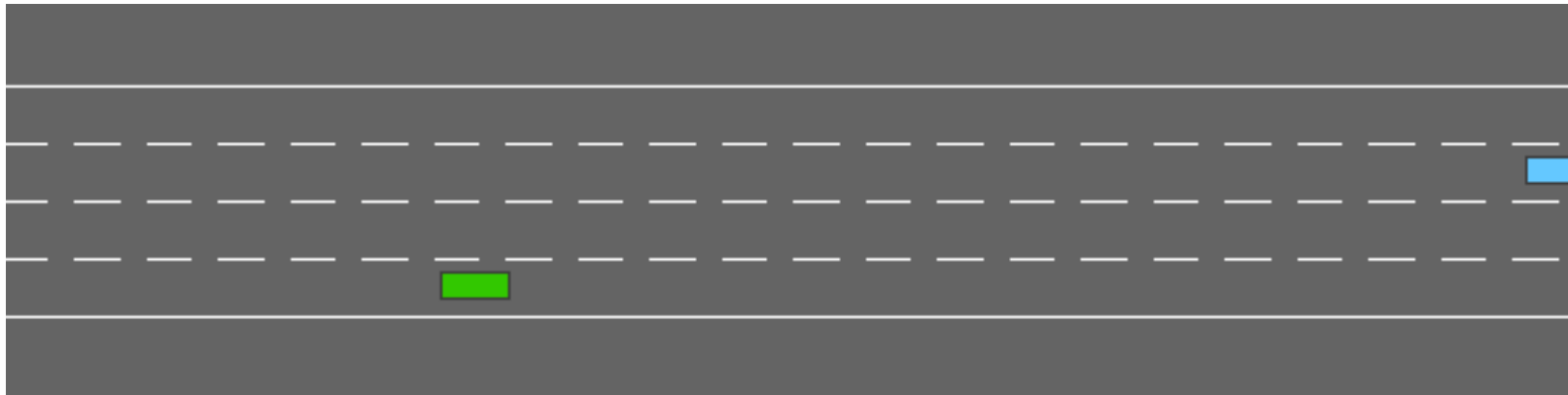
Why is this so difficult in practice?

- Uncertainty, robustness, humans, ...
- Worlds like Go, chess follow simple rules (although succeeding in these games is certainly not simple ...).
 - Closed-world assumption holds true.
- CWA not true for real driving/flying environments; Anything can happen
- Modeling the real world is difficult; significant model errors inevitable
- How do you ensure safety? Humans are fragile and unpredictable.
- A main challenge — introduce robustness, safety, and resilience

Most likely, performant systems will consist of learning systems and advanced control, (and ...)

HI – Learning and autonomous vehicles

- The HI for learning in this course is aimed to get initial understanding of two basic techniques
- Basic HI
 1. Basic RL understanding, small academic example
 2. Neural networks for behavior prediction
- Extra assignment — Deep Q-learning for (reckless) highway driving

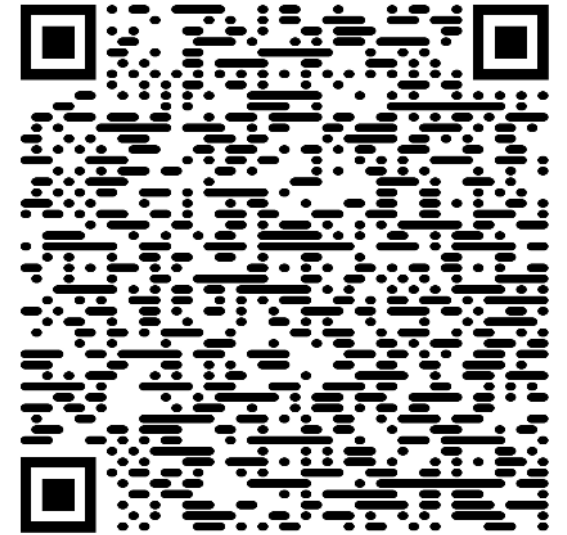


CERBERUS Deployment at STIX 2019

DARPA Subterranean Challenge



CERBERUS



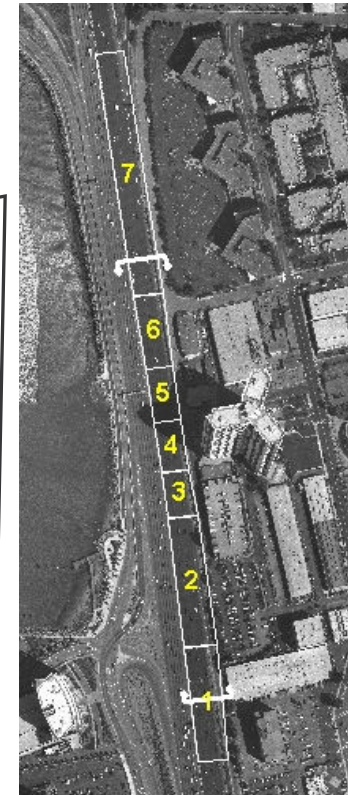
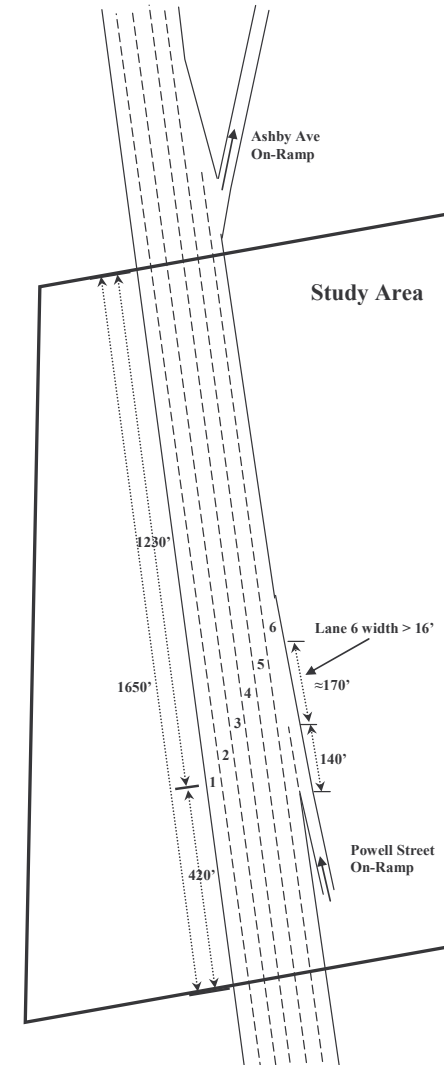
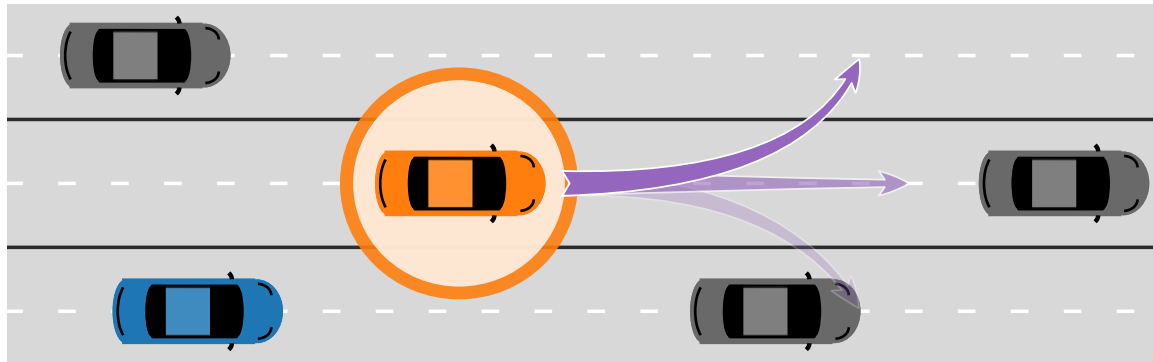
<https://rsl.ethz.ch/research/challenges-competitions/cerberus.html>



A possible use-case - driver prediction

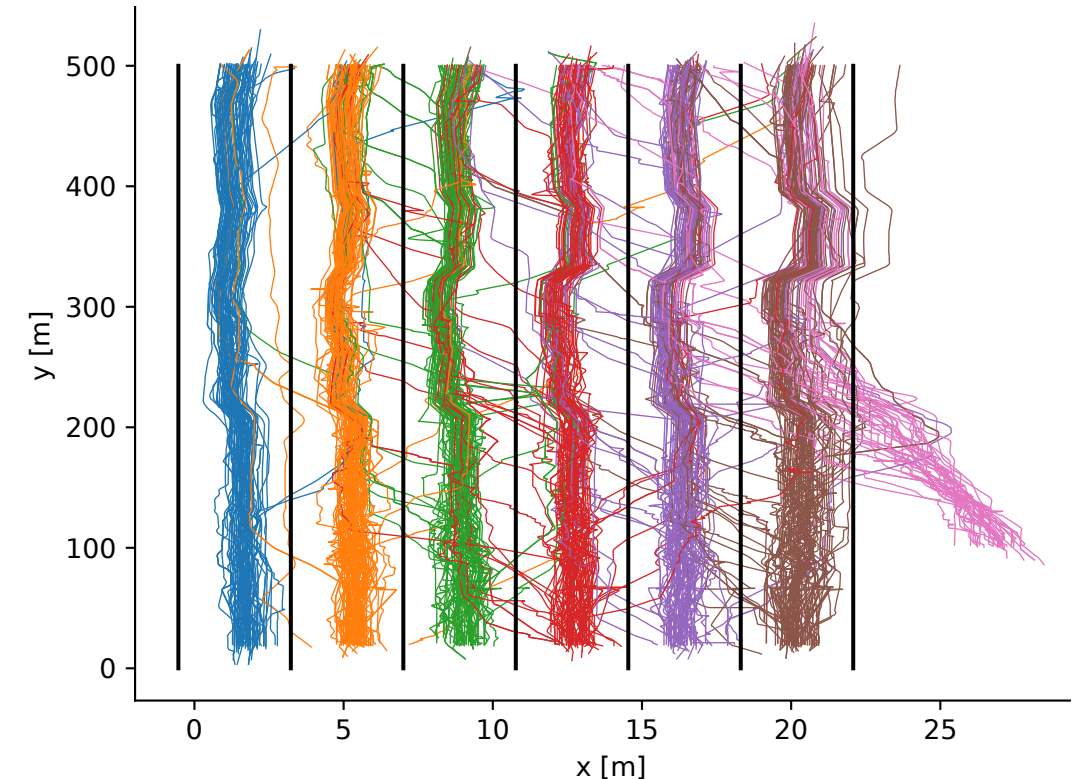
Example - behavioral models of drivers

- Physical models are very useful, they can be understood, and they extrapolate well.
- We know how to model basic mechanics; but how do we model human behavior?
- Consider the 6 lane highway, a section of the I-80 in Emeryville outside Oakland
- To plan safe motion in high density traffic, it would be good to have predictions what surrounding drivers will do next



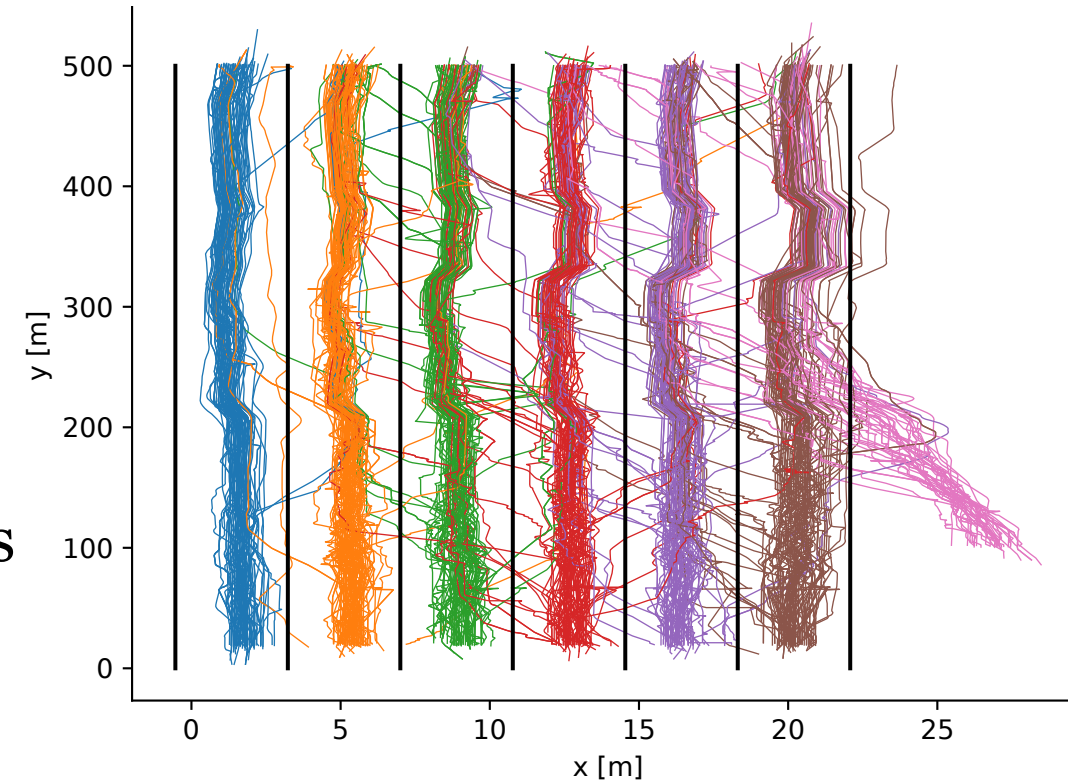
What could influence a driver? (1/2)

- The (unknown) long-term plan
- Surrounding traffic - positions, velocities, accelerations
- Many aim for high speed lanes to the left, i.e., lane shifts to the left more common than the opposite
- Could depend on mean velocity and density, in lanes
- ... and then some



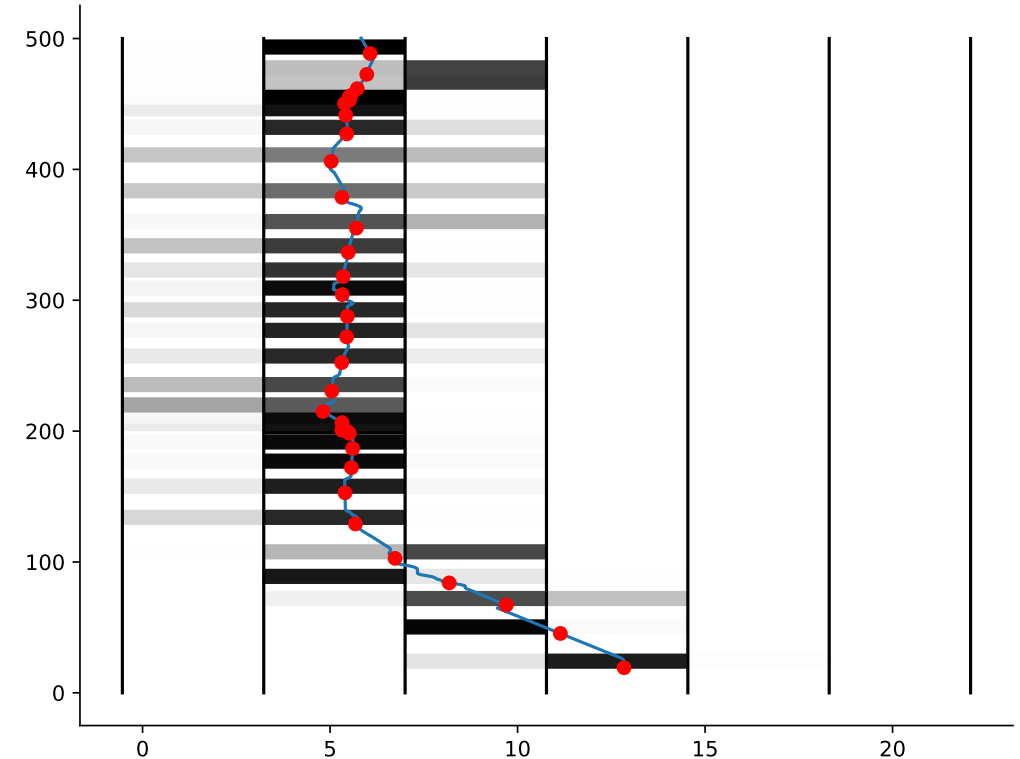
What could influence a driver? (2/2)

- Predicting what a driver will do short-term (less than a second), look at the velocity vector
- On a medium time-scale 1-3 seconds, drivers start to interact and act with the environment
- Long-term, then strategic driver plans
- Building a rule-based model for driver behavior will be difficult

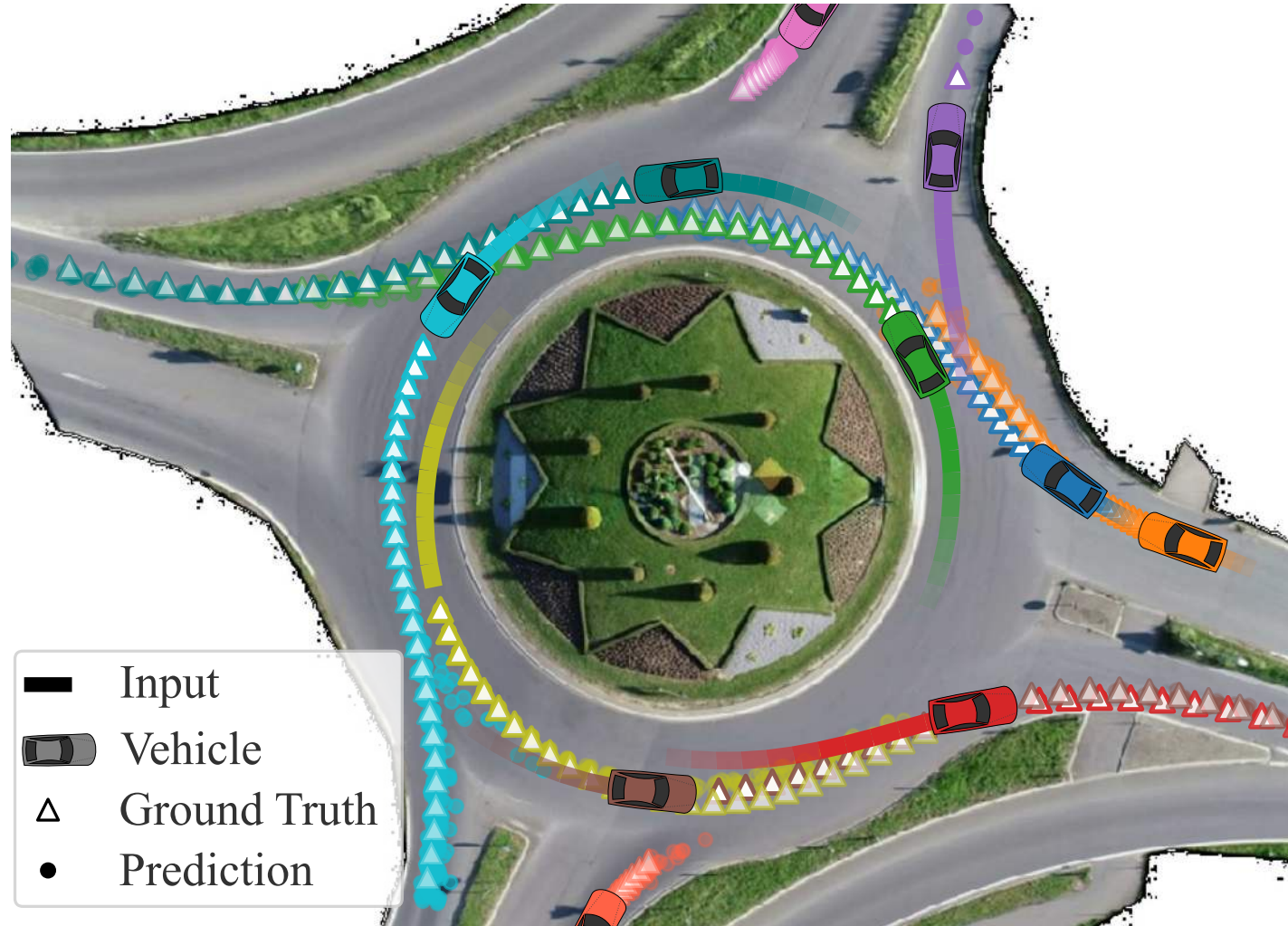


A simple approach

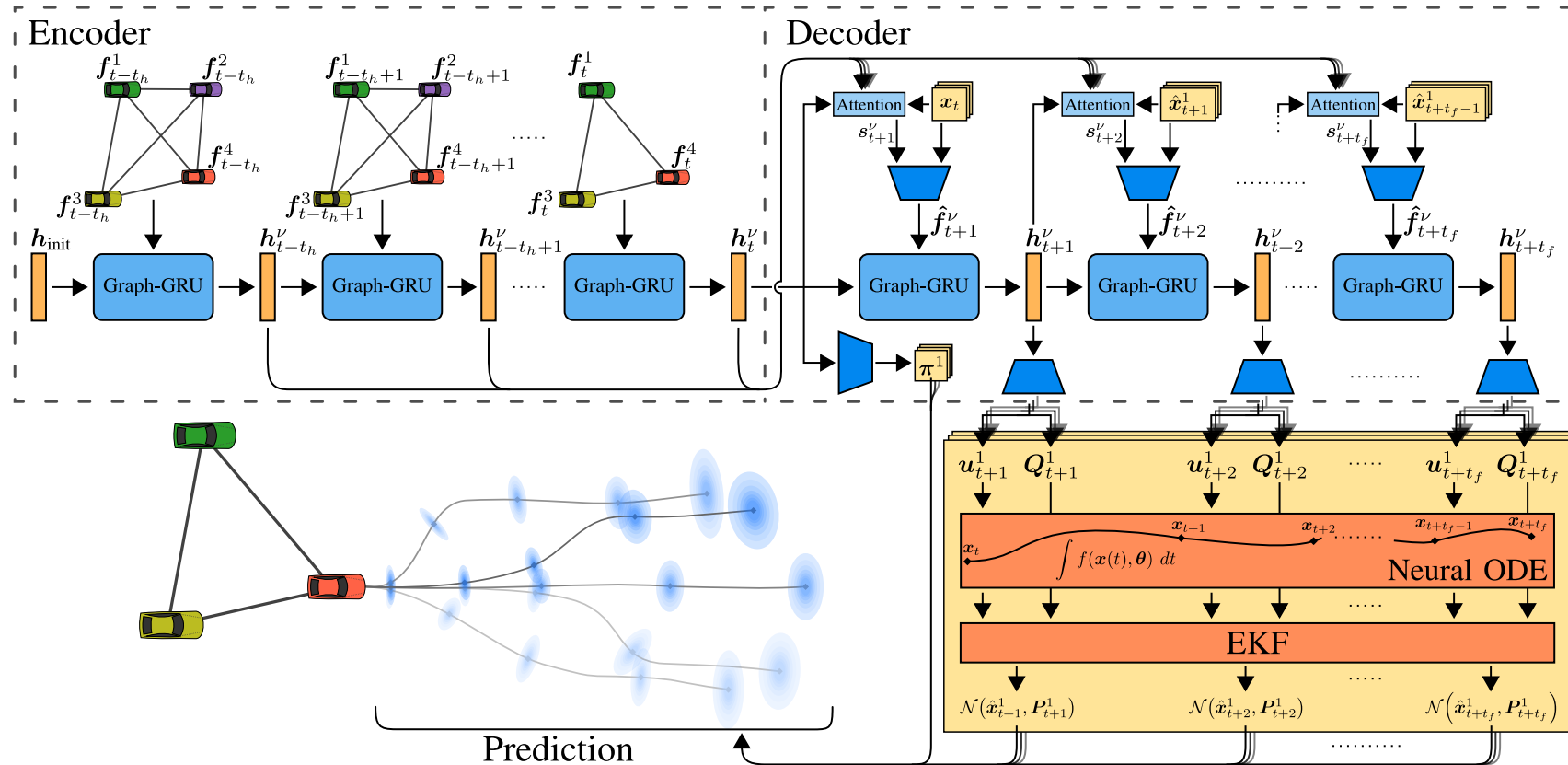
- Maybe using recorded data to build ML models can be used to predict driving behavior
- Find features x , data about the current situation and try to predict if a driver will change lanes within the next 3 seconds
- Model
$$\hat{y} = f(x) \in \{\text{left, stay, right}\}$$
- What is a good feature vector x ?



A more state-of-the-art approach (1/2)



A more state-of-the-art approach (2/2)



$\approx 200.000 - 300.000$ parameters (not small but not big)

STATE
Autonomous

MPH
3



5x ▶▶



Kyle Vogt



Sam Altman



<https://youtu.be/sliYTyRpRB8?si=wEohMsZ7jbPragy0>

Summary of introduction

- Learning for autonomous vehicles is still very much an active research area
- Computer vision, detection and classification of objects in images, semantic segmentation of the world
 - areas where learning techniques are already important and essential
- We are dealing with mechatronic systems, classical control will be important
 - Model Predictive Control is an especially interesting branch of control
- A basic research question for learning and control;
how, where, and when they can collaborate to realize safe, robust, and resilient autonomous systems

Why should you become a PhD student?

An information event in Visionen on November 12, 17:00

The aim of the evening is to give some answers to

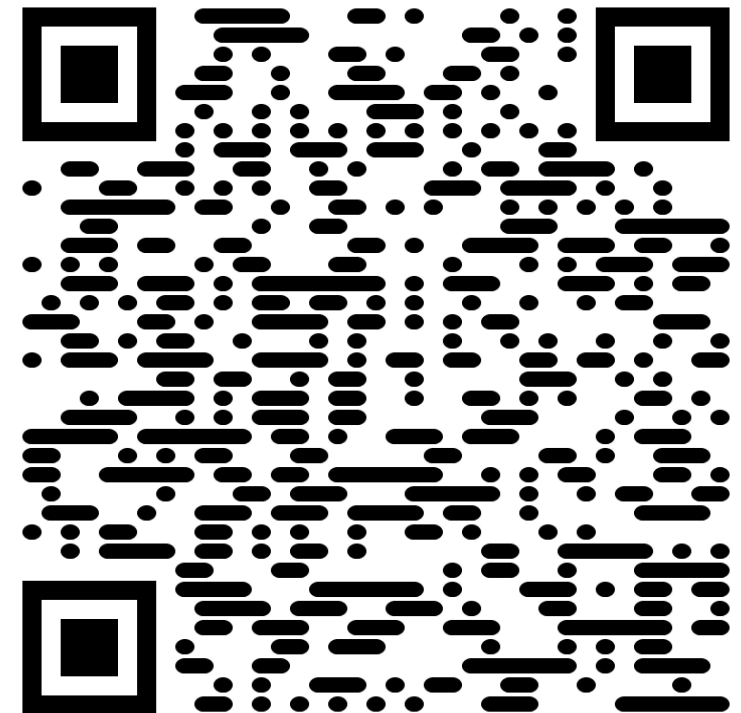
- Why should you become a PhD student?
- How does it happen?
- What can you research in?

During the evening you will have the opportunity to listen and talk with both current PhD students and people who have completed their PhDs and now works in industry.

Where: ISY's arena Visionen in B-huset, and we offer sandwiches and drinks to registered participants.

<https://forms.office.com/e/BX4siQAaCk>

Registration deadline October 31



www.liu.se